

SOFTWARE REQUIREMENTS SPECIFICATION

Rental Property Management System (RPMS)

Course Name: Software Testing and Quality Assurance

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Group Number: 3

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Software Requirements Specification

RentalHub / House Rental System

Field	Value
Document ID	RH-SRS-2026-001
System	RentalHub / House Rental System
Version	1.1
Status	Updated for current reservation workflow
Prepared for	Project stakeholders and development team
Prepared on	June 19, 2026
Standard basis	ISO/IEC/IEEE 29148-inspired SRS structure

Prepared from project documentation, backend controllers and services, entities, migrations, and frontend routes/components.

1. Introduction

1.1 Purpose

This SRS defines the functional, data, interface, security, quality, and operational requirements for RentalHub, a web-based house rental platform for property discovery, landlord onboarding, single-property reservation holds, viewing appointment confirmation, property management, reviews, notifications, and administrative oversight.

1.2 Scope

- Public users can browse listings, view property details, register, and submit landlord onboarding requests.
- Tenants can reserve available properties through a single 24-hour hold, request an exact viewing appointment before move-in, manage reservations, favorite listings, and submit verified reviews.
- Landlords can manage owned properties, confirm requested viewing appointments, cancel requests they cannot support, and accept confirmed tenant reservations.
- Agents can process assigned and locality-relevant landlord onboarding or additional-property verification requests and manage verification documents.
- Admins can manage users, review platform data, approve properties, inspect requests, and view audit logs.

1.3 Definitions

Term	Definition
FCFS	Legacy term for first-come-first-served queueing; the current workflow uses one active reservation hold per property instead of a queue.
OTP	One-time passcode sent by email for verification.
JWT	JSON Web Token used for stateless API authentication.
Tenant	Authenticated user who searches for rentals, places reservation holds, requests viewing appointments, manages reservations, and submits eligible reviews.
Landlord	Verified property owner or host who manages listings, confirms requested viewing appointments, and accepts confirmed reservations.
Agent	Operational role that verifies landlord requests and documents.
Admin	System role with user, property, request, reservation, and log oversight.

2. Overall Description

2.1 Product Perspective

RentalHub is a full-stack application built with a React/Vite frontend and a Spring Boot REST API backed by PostgreSQL. It can run as separate development services or serve the built SPA through the Spring Boot backend.

2.2 User Classes

User class	Primary responsibilities
Guest	Browse marketplace, view details, register, request landlord onboarding, read manual/about pages.
Tenant	Browse, reserve one property hold at a time per property, request viewing appointment date/time, track holds/rentals, favorite properties, submit eligible reviews.
Landlord	Manage owned listings, keep available listings serviceable for viewing, review landlord-side reservations, confirm requested appointments, cancel unsupported requests, accept confirmed reservations.
Agent	Review assigned and locality-visible landlord/property requests, upload documents, approve/reject requests, inspect locality properties.
Admin	Manage users, approve properties, inspect all reservations/bookings/logs/requests, view statistics.

2.3 Operating Environment

- Backend: Java 17+, Spring Boot 3.x, Spring Security, Spring Data JPA, Flyway, PostgreSQL.
- Frontend: React 18, Vite, React Router, Axios, Tailwind CSS, Framer Motion, Lucide React.
- External services: SMTP email, Cloudinary-compatible image storage, local upload directory for documents/files.
- Default backend port: 8080. Default frontend development port: 5173.

2.4 Constraints, Assumptions, and Dependencies

- Database schema is managed by Flyway migrations.
- Email verification data is currently held in memory and may not survive server restarts.
- JWT token lifetime defaults to 24 hours unless configured otherwise.
- Cloudinary and mail credentials must be configured for production-grade image and email behavior.
- Legacy booking endpoints exist, but the implemented primary rental workflow is the reservation hold and landlord appointment confirmation process.

3. External Interface Requirements

3.1 User Interface

- Public pages: home, about, manual, login, register, marketplace, property detail, landlord join request, landlord email verification.
- Protected pages: dashboard, tenant reservations, tenant rentals, property form, property edit form.
- Reusable UI: navbar, footer, notification bell, OTP modal, document uploader, reservation hold status, reservation calendar, viewing appointment input, and cost estimator.

3.2 API Interface

Area	Representative endpoints
Authentication	/api/v1/auth/register, /verify, /login, /logout, /me, /profile, /resend-verification-code
Properties	/api/v1/properties, /my, /my-locality, /{id}, /{id}/approve, /{id}/images
Reservations	/api/v1/reservations, /my, /landlord, /property/{id}/queue (public hold status), /property/{id}/available-dates, /{id}/confirm-appointment, /{id}/confirm, /{id}/accept, /{id}/cancel, /{id}/confirmation-letter
Landlord requests	/api/v1/landlord-requests, /additional-property, /verify, /verify-and-activate, /my, /{id}/assign, /{id}/documents, /{id}/approve, /{id}/reject
Notifications	/api/v1/notifications, /unread-count, /{id}/read, /read-all
Reviews	/api/v1/properties/{propertyId}/reviews, /eligibility
Administration	/api/v1/admin/users, /stats, /bookings, /reservations, /logs

3.3 Data and Storage Interfaces

- PostgreSQL stores users, properties, reservations, notifications, landlord requests, documents, reviews, favorites, bookings, messages, and audit logs.
- Multipart uploads are used for property images and landlord verification documents.
- PDF output is used for rental confirmation letters.

4. Functional Requirements

The following catalog lists the discovered functional requirements from documentation, routes, controllers, DTOs, entities, and frontend integrations.

ID	Category	Requirement	Priority
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FR-AUTH-01	Authentication	The system shall allow public users to register tenant accounts using full name, email, password, and role information.	High
FR-AUTH-02	Authentication	The system shall queue registration and send a six-digit OTP email before account creation.	High
FR-AUTH-03	Authentication	The system shall verify submitted OTP codes before creating a user account.	High
FR-AUTH-04	Authentication	The system shall allow users to resend registration verification codes.	Medium
FR-AUTH-05	Authentication	The system shall authenticate users by email and password and return a JWT bearer token with user details.	High
FR-AUTH-06	Authentication	The system shall expose the current authenticated user profile.	High
FR-AUTH-07	Authentication	The system shall allow authenticated users to update profile details.	Medium
FR-AUTH-08	Authentication	The system shall support logout activity for session clearing and audit logging.	Medium
FR-AUTH-09	Authentication	The system shall reject inactive accounts and invalid credentials.	High
FR-AUTH-10	Authentication	The frontend shall persist JWT tokens locally and attach them to authenticated API requests.	High
FR-RBAC-01	Access control	The system shall enforce role-based access for tenant, landlord, agent, and admin roles.	High
FR-RBAC-02	Access control	The system shall redirect unauthenticated users away from protected frontend routes.	High
FR-RBAC-03	Access control	The system shall redirect authenticated users away from guest-only login and registration pages.	Medium
FR-RBAC-04	Access control	The system shall limit dashboard access to landlord, agent, and admin users.	High
FR-RBAC-05	Access control	The system shall limit tenant reservation and rental pages to tenant users.	High
FR-RBAC-06	Access control	The system shall limit property creation to admin and agent users in the current frontend route configuration.	High

FR-RBAC-07	Access control	The system shall permit property editing to property owners, admins, and agents according to backend authorization.	High
FR-PROP-01	Properties	The system shall display public property listings with pagination.	High
FR-PROP-02	Properties	The system shall allow filtering listings by location.	High
FR-PROP-03	Properties	The system shall allow filtering listings by maximum monthly price.	High
FR-PROP-04	Properties	The system shall allow filtering listings by minimum room count.	High
FR-PROP-05	Properties	The system shall allow filtering listings by availability status.	High
FR-PROP-06	Properties	The system shall display detailed property information including title, description, location, price, room count, availability, images, and contact data where authorized.	High
FR-PROP-07	Properties	The system shall allow authorized users to create property listings.	High
FR-PROP-08	Properties	The system shall allow authorized owners, agents, and admins to update property listings.	High
FR-PROP-09	Properties	The system shall allow authorized owners, agents, and admins to delete property listings.	High
FR-PROP-10	Properties	The system shall allow authorized users to upload one or more property images using multipart upload.	High
FR-PROP-11	Properties	The system shall store and return property image URLs or file paths.	High
FR-PROP-12	Properties	The system shall support admin approval or rejection of property listings.	High
FR-PROP-13	Properties	The system shall list properties owned by the current user.	High
FR-PROP-14	Properties	The system shall list properties in an agent's locality for locality-based review or assignment.	Medium
FR-PROP-15	Properties	The system shall hide unapproved properties from unauthorized public viewing.	High

FR-SEARCH-01	Search	The system shall provide a public search endpoint for property discovery.	Medium
FR-SEARCH-02	Search	The frontend shall provide marketplace grid and list-style browsing experiences.	Medium
FR-RES-01	Reservations	The system shall allow authenticated tenants to create reservation holds for approved and available properties.	High
FR-RES-02	Reservations	The system shall allow only one active pre-rental reservation hold for a property at a time.	High
FR-RES-03	Reservations	The system shall expose public reserved status, reservation status, hold expiry time, and appointment confirmation state without exposing tenant personal details.	High
FR-RES-04	Reservations	The system shall calculate and expose the earliest available move-in date for each property based on active rentals and lease schedule.	High
FR-RES-05	Reservations	The system shall allow tenants to view their own reservations, hold state, appointment date/time, move-in date, duration, and estimated cost.	High
FR-RES-06	Reservations	The system shall allow landlords to view reservations and requested viewing appointments for their properties.	High
FR-RES-07	Reservations	The system shall create a 24-hour exclusive hold when a tenant reserves a property.	High
FR-RES-08	Reservations	The system shall require the tenant to select an exact viewing appointment date and time before completing a reservation request.	High
FR-RES-09	Reservations	The system shall reject reservation requests when the requested viewing appointment is not before the move-in date.	High
FR-RES-10	Reservations	The system shall allow landlords to confirm requested viewing appointments directly in the system.	High
FR-RES-11	Reservations	The system shall allow authenticated tenants and landlords to cancel reservations they are permitted to cancel.	High
FR-RES-12	Reservations	The system shall expire active reservation holds that pass the 24-hour deadline without landlord appointment confirmation.	High

FR-RES-13	Reservations	The system shall release a property for other tenants after the active hold is cancelled, declined, or expired.	High
FR-RES-14	Reservations	The system shall prevent duplicate active reservations for the same tenant and property.	High
FR-RES-15	Reservations	The system shall compute estimated total cost from monthly rent and duration.	Medium
FR-RES-16	Reservations	The system shall generate downloadable PDF confirmation letters for authorized reservation users after landlord acceptance.	Medium
FR-RES-17	Reservations	The tenant frontend shall show reservation hold status, landlord appointment confirmation, hold expiry, move-in details, and available cancellation actions.	High
FR-RES-18	Reservations	The property detail page shall support move-in date selection, exact viewing appointment date/time selection, duration-based cost estimation, and public hold status display.	High
FR-LAND-01	Landlord onboarding	The system shall allow public users to submit landlord join requests.	High
FR-LAND-02	Landlord onboarding	Landlord join requests shall collect applicant identity, contact, locality, TIN, and property registration data.	High
FR-LAND-03	Landlord onboarding	Existing landlords shall be able to submit additional-property requests.	High
FR-LAND-04	Landlord onboarding	The system shall support landlord request statuses of pending, assigned, verified, approved, and rejected.	High
FR-LAND-05	Landlord onboarding	Agents and admins shall be able to assign landlord requests to agents.	High
FR-LAND-06	Landlord onboarding	Agents shall be able to view requests assigned to them and landlord/property verification requests in their locality.	High
FR-LAND-07	Landlord onboarding	Admins shall be able to view all landlord requests.	High
FR-LAND-08	Landlord onboarding	Agents and admins shall be able to upload verification documents against landlord requests and request properties.	High
FR-LAND-09	Landlord onboarding	Agents and admins shall be able to approve landlord requests.	High

FR-LAND-10	Landlord onboarding	Agents and admins shall be able to reject landlord requests with a reason.	High
FR-LAND-11	Landlord onboarding	The system shall send landlord email verification messages and activation OTP codes.	High
FR-LAND-12	Landlord onboarding	Approved landlords shall activate accounts by verifying email and setting a password.	High
FR-LAND-13	Landlord onboarding	Agents and admins shall be able to resend landlord verification emails.	Medium
FR-NOT-01	Notifications	The system shall create in-app notifications for relevant reservation holds, viewing appointments, reservation cancellations/expiry, landlord request, property approval, review, and general events.	High
FR-NOT-02	Notifications	The system shall send descriptive email notifications with HTML bodies where email integration is configured.	Medium
FR-NOT-03	Notifications	Authenticated users shall be able to view their notifications ordered by creation time.	High
FR-NOT-04	Notifications	Authenticated users shall be able to retrieve unread notification counts.	High
FR-NOT-05	Notifications	Authenticated users shall be able to mark a notification as read.	Medium
FR-NOT-06	Notifications	Authenticated users shall be able to mark all notifications as read.	Medium
FR-NOT-07	Notifications	The frontend shall poll notifications periodically and show unread badges.	Medium
FR-REV-01	Reviews	The system shall display public property reviews.	Medium
FR-REV-02	Reviews	The system shall determine whether the current tenant is eligible to review a property.	High
FR-REV-03	Reviews	The system shall allow eligible tenants to submit ratings and review comments.	High
FR-REV-04	Reviews	The system shall enforce tenant-only verified review creation based on completed or accepted rental history.	High
FR-FAV-01	Favorites	The system shall allow authenticated tenants to list favorite properties.	Medium

FR-FAV-02	Favorites	The system shall allow authenticated tenants to add a property to favorites.	Medium
FR-FAV-03	Favorites	The system shall allow authenticated tenants to remove a property from favorites.	Medium
FR-BOOK-01	Legacy bookings	The system shall retain legacy booking endpoints for tenant booking creation and status updates.	Low
FR-BOOK-02	Legacy bookings	The system shall allow tenants, landlords, and admins to retrieve bookings according to role.	Low
FR-ADMIN-01	Administration	Admins shall be able to list all users.	High
FR-ADMIN-02	Administration	Admins shall be able to create users.	High
FR-ADMIN-03	Administration	Admins shall be able to update users.	High
FR-ADMIN-04	Administration	Admins shall be able to patch user account attributes such as active state and role.	High
FR-ADMIN-05	Administration	Admins shall be able to delete users.	High
FR-ADMIN-06	Administration	Admins shall be able to retrieve global platform statistics.	High
FR-ADMIN-07	Administration	Admins shall be able to retrieve all bookings.	Medium
FR-ADMIN-08	Administration	Admins shall be able to retrieve all reservations.	High
FR-ADMIN-09	Administration	Admins shall be able to retrieve system audit logs.	High

FR-DASH-01	Dashboards	The system shall provide role-specific dashboard content for admins, agents, landlords, and tenants.	High
FR-DASH-02	Dashboards	The landlord dashboard shall display owned listings, reservation requests, and portfolio metrics.	High
FR-DASH-03	Dashboards	The admin dashboard shall display users, properties, reservations, logs, and landlord requests.	High
FR-DASH-04	Dashboards	The agent dashboard shall display assigned landlord requests, locality-visible landlord/property verification requests, and locality-relevant properties.	High
FR-DASH-05	Dashboards	The tenant dashboard and tenant pages shall display reservation hold state, appointment status, rentals, and downloadable confirmation letters.	High
FR-UI-01	User interface	The system shall provide public pages for home, about, manual, login, register, property browsing, property detail, landlord request, and landlord verification.	Medium
FR-UI-02	User interface	The system shall provide responsive navigation, footer, dark mode support, and mobile menu behavior.	Medium
FR-UI-03	User interface	The system shall provide user-facing validation and toast-style feedback for major workflows.	Medium
FR-UI-04	User interface	The system shall provide a manual page for user guidance.	Low
FR-LOG-01	Audit logging	The system shall log critical authentication, registration, property, reservation, notification, and administration events.	High
FR-LOG-02	Audit logging	Audit log entries shall include action, entity type, entity id, user identity, details, and timestamp where available.	High
FR-RES-19	Reservations	The system shall limit a tenant to a maximum of three active reservation holds or confirmed pre-rental reservations at a time.	High
FR-RES-20	Reservations	The landlord dashboard shall instruct landlords to keep online available properties serviceable for viewing appointments or cancel requests they cannot attend.	Medium

5. Nonfunctional Requirements

ID	Category	Requirement	Priority
NFR-SEC-01	Security	The system shall use stateless JWT authentication for protected API access.	High

NFR-SEC-02	Security	The system shall hash stored passwords using a secure password encoder.	High
NFR-SEC-03	Security	The system shall enforce CORS using configured allowed origins.	High
NFR-SEC-04	Security	The system shall restrict sensitive landlord documents and admin functions to authorized roles.	High
NFR-SEC-05	Security	JWT secrets and external service credentials shall be supplied through environment variables or secure configuration.	High
NFR-PRIV-01	Privacy	The system shall avoid exposing private contact and verification data to unauthorized public users.	High
NFR-PERF-01	Performance	Common list endpoints shall support pagination or bounded retrieval to reduce response payload size.	Medium
NFR-PERF-02	Performance	Notification and reservation hold-status polling intervals shall be short enough to keep the UI current without excessive load.	Medium
NFR-REL-01	Reliability	Reservation status transitions shall be transactional so active hold exclusivity, status, notifications, and property state remain consistent.	High
NFR-REL-02	Reliability	Email failures shall be logged without corrupting the associated business transaction.	Medium
NFR-REL-03	Reliability	Scheduled expiry shall run automatically to release reservation holds that pass the 24-hour deadline.	High
NFR-MAINT-01	Maintainability	Backend code shall separate controllers, services, repositories, DTOs, entities, security, and configuration.	Medium
NFR-MAINT-02	Maintainability	Database changes shall be managed through Flyway migrations.	High
NFR-MAINT-03	Maintainability	Frontend code shall separate pages, reusable components, context, and utilities.	Medium
NFR-COMP-01	Compatibility	The frontend shall run in modern browsers and be built with React and Vite.	Medium
NFR-COMP-02	Compatibility	The backend shall run on Java 17+ with Spring Boot 3.x.	High

NFR-DATA-01	Data integrity	The system shall enforce unique user email addresses.	High
NFR-DATA-02	Data integrity	The system shall enforce valid reservation and landlord request status values through enumerations.	High
NFR-USAB-01	Usability	Primary workflows shall provide clear status, loading, success, and error feedback.	Medium
NFR-USAB-02	Usability	The property search and reservation hold flows shall be usable from desktop and mobile layouts.	Medium
NFR-SCAL-01	Scalability	Image storage shall support external hosting through Cloudinary or equivalent configured storage.	Medium
NFR-PORT-01	Portability	The application shall be deployable as separate frontend/backend services or as a unified Spring Boot-served SPA build.	Medium

6. Data Requirements

Entity	Required data
User	Email, password hash, role, full name, active state, email verification state, locality, phone, TIN, creator, created timestamp.
Property	Landlord, title, description, location, monthly price, rooms, availability, approval state, contact phone/email, image records.
Reservation	Property, tenant, compatibility queue position, status, move-in date, viewing appointment date/time, duration, estimated cost, 24-hour hold deadline, appointment confirmation timestamp, landlord response notes, timestamps.
Landlord request	Requester identity/contact, locality, TIN, status, type, assigned agent, created landlord, notes, property claims.
Document	Landlord request, document type, optional request property, file path, uploader, timestamps.
Notification	Recipient, type, title, message, read flag, email-sent flag, reference id, creation time; reservation emails should support descriptive HTML bodies.
Review	Property, tenant, rating, comment, timestamps; tenant eligibility must be verified.
System log	Action, entity type/id, user id/email, details, creation time.

7. Use Case Summary

ID	Use case	Actors	Summary
UC-01	Register and verify tenant account	Guest	Guest submits registration; system emails OTP; guest verifies OTP; account is created.
UC-02	Browse and filter properties	Guest or authenticated user	User opens marketplace, applies filters, views paged listings, and opens property details.
UC-03	Reserve property and request viewing appointment	Tenant	Tenant selects move-in date, duration, and exact viewing appointment time; system creates a single 24-hour hold if the property and tenant limits allow it.
UC-04	Confirm appointment and accept reservation	Landlord	Landlord reviews the requested viewing appointment, confirms it if available, then may accept the confirmed reservation; cancellation or expiry releases the property hold.
UC-05	Submit landlord onboarding request	Prospective landlord	Applicant submits identity, TIN, locality, contact, and property data for review.
UC-06	Review landlord request	Agent or Admin	Agent/admin assigns, uploads verification documents, approves or rejects, and sends verification email.
UC-07	Manage platform users	Admin	Admin lists users, creates or edits users, changes active state or role, and deletes accounts.
UC-08	Review property after rental	Tenant	Tenant with eligible rental history opens property detail and submits rating/comment.

8. Requirement Traceability

Feature area	Requirement IDs
Authentication and RBAC	FR-AUTH-01..10, FR-RBAC-01..07, NFR-SEC-01..05
Property marketplace and management	FR-PROP-01..15, FR-SEARCH-01..02, FR-UI-01..03

Reservation holds, appointments, and rentals	FR-RES-01..20, NFR-REL-01..03
Landlord onboarding and agent work	FR-LAND-01..13, FR-DASH-04
Notifications	FR-NOT-01..07, reservation HTML email behavior
Reviews and favorites	FR-REV-01..04, FR-FAV-01..03
Administration and audit	FR-ADMIN-01..09, FR-LOG-01..02
Legacy booking compatibility	FR-BOOK-01..02

9. Open Issues and Recommendations

- Decide whether legacy booking endpoints should remain in scope or be retired in favor of the reservation hold workflow.
- Persist OTP verification state in a database table or external cache for production reliability.
- Add refresh-token or session-renewal behavior if long-running sessions are required.
- Clarify whether landlords should be allowed to create new listings directly from the frontend after onboarding, because backend capability and frontend route restrictions are not perfectly aligned.
- Define production file retention, document privacy, and Cloudinary deletion policies.